Solutions

SPEECH PROCESSING

1. A spectrogram is defined at time-frame m and frequency index k by

$$X(k,m) = \sum_{r=0}^{N-1} w(r)x(m-r)\exp\left(-\frac{j2\pi k(m-r)}{N}\right)$$

where the input signal is

$$x(n) = \begin{cases} 0 & \text{for } n < 0 \\ \exp\left(\frac{j2\pi fn}{N}\right) & \text{for } n \geq 0 \end{cases}$$

and the window function with length $R \leq N$ is

$$w(r) = \begin{cases} \frac{1}{R} & \text{for } r < R \\ 0 & \text{for } r \geq R. \end{cases}$$

a) i) Draw a fully labelled sketch of the window function. [2]

ii) Consider denoting the upper limit on the summation in the above equation for the spectrogram as $Q-1$. Determine Q for which the terms of the summation are non-zero. [2]

iii) For the case of $0 \leq m < R$, show that [4]

$$|X(k,m)| = \left|\frac{\sin\left(\frac{\pi(f-k)(m+1)}{N}\right)}{R\sin\left(\frac{\pi(f-k)}{N}\right)}\right|.$$

iv) For the case of $m \geq R$, find an expression for $|X(k,m)|$. [2]

The following identities can be used:

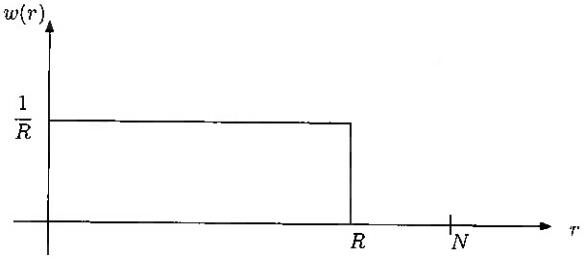
$$\sum_{r=0}^{N-1} \exp(j2\pi(ar+b)) = \frac{\sin(N\pi a)}{\sin(\pi a)}\exp(j\pi(a(N-1)+2b)),$$

$$\frac{\sin(ax)}{\sin(bx)} \xrightarrow{x\to 0} \frac{a}{b}.$$

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Solution:

Window function:



Outside the range $(m-r) \geq 0$ and $r < R$ either x or w are zero and hence the limit for the summation can be set at $Q-1$ where $Q = \min(m+1, R)$. Expanding gives

$$X(k,m) = \sum_{r=0}^{Q-1} \frac{1}{R} \exp\left(\frac{j2\pi f(m-r)}{N}\right) \exp\left(\frac{-j2\pi k(m-r)}{N}\right)$$
$$= \frac{1}{R} \sum_{r=0}^{Q-1} \exp\left(j2\pi \frac{(f-k)m-(f-k)r}{N}\right)$$

In order to use the identities given, expressions for a and b are found from comparing terms

$$ar+b = \frac{(f-k)m-(f-k)r}{N}$$

from which

$$a = \frac{-(f-k)}{N} \quad b = \frac{(f-k)m}{N}$$

and so

$$X(k,m) = \frac{1}{R} \exp\left(j\pi \frac{2(f-k)m-(f-k)(Q-1)}{N}\right) \frac{\sin\left(\frac{Q\pi(f-k)}{N}\right)}{\sin\left(\frac{\pi(f-k)}{N}\right)}.$$

The modulus of the exponential is unity and hence

$$|X(k,m)| = \left|\frac{\sin\frac{Q\pi(f-k)}{N}}{R\sin\frac{\pi(f-k)}{N}}\right|.$$

When $0 \leq m < R$, it follows that $Q = m+1$ and hence the result is shown. When $m \leq R$ then $Q = R$ leading to

$$|X(k,m)| = \left|\frac{\sin\frac{R\pi(f-k)}{N}}{R\sin\frac{\pi(f-k)}{N}}\right|.$$

b) Consider the case of $k = f$ and varying m, with $N = 128$. [4]

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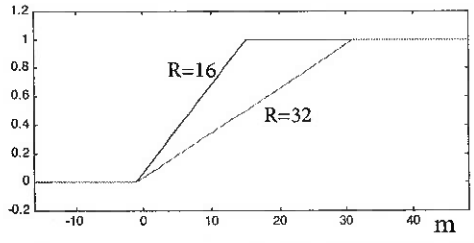
i) Determine expressions for $|X(k,m)|$ for the ranges of m:

- $m < 0$
- $0 \leq m < R$
- $m \geq R$.

ii) Draw labelled sketch graphs of $|X(k,m)|$ as a function of m for $R = 16$ and $R = 32$. Include both graphs on the same axes and cover the range $-16 \leq m \leq 48$.

Solution: In the case of $k = f$, the expression gives

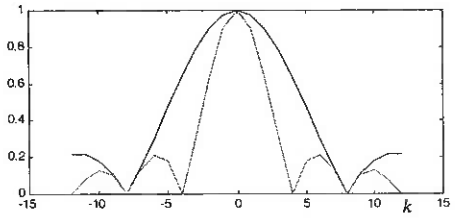
$$|X(k,m)| = \begin{cases} 0 & m < 0 \\ \frac{m+1}{R} & 0 \leq m < R \\ 1 & m \geq R. \end{cases}$$



c) Consider the case of $m = 127$ and varying k, with $N = 128$. [3]

Draw labelled sketch graphs of $|X(k,m)|$ as a function of k for $R = 16$ and $R = 32$. Include both graphs on the same axes and cover the range $f - 12 \leq k \leq f + 12$

Solution:



d) Explain how the choice of R in spectrographic analysis affects the time and frequency resolution. Relate your explanation to your answers in parts (b) and (c). [3]

Solution: The longer the effective window length, R, the better the frequency resolution but the worse the time resolution. The input signal here begins only at t=0 and is a complex cosine wave. It is shown in part (b) that the longer of the two windows takes twice as long to respond to the start of the input signal. In contrast, in part (c), it is seen that the longer of the windows has a main lobe in the frequency domain with 50% of the width.

2. a) Consider a speech recognizer based on a left-to-right, no-skips Hidden Markov Model. Consider a particular state of this recognizer. The probability of remaining in the state from one frame to the next frame is $(1-p)$.

i) Explain why the state sequence is described as a Markov process. [1]

ii) Prove that the number of frames, D, aligned to this state has a negative exponential distribution with an average of $\frac{1}{p}$. [5]

Solution: The probability of being in any state for the next frame depends only on the current state and not on any previous history, hence it is a Markov process.
There will always be $(n-1)$ self transitions plus one forward transition. Hence,

$$pr(D=n) = p(1-p)^{n-1}.$$

Taking the expectation gives

$$\mathscr{E}(D) = \sum_{n=1}^{\infty} np(1-p)^{n-1} = \frac{1}{p}$$

since by differentiating

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \Longrightarrow \sum_{n=0}^{\infty} nx^{n-1} = \frac{1}{(1-x)^2}.$$

b) Define and explain the Fisher Ratio and its relevance to speech recognition. Include in your answer [4]

i) a conceptual definition with any appropriate diagrams,

ii) a straightforward mathematical definition,

iii) a definition in terms of appropriately defined matrices,

iv) an explanation of its relevance to speech recognition.

Solution: The Fisher Ratio is defined as the ratio of the variance of the class means to the average variance within a class. It is relevant to speech recognition because speech recognition performs classification. More specifically, the Fisher Ratio enables the discriminate capability of any particular feature set, or transformed feature set, to be evaluated. In terms of matrices, and considering $\mathbf{W}$ and $\mathbf{B}$ as the average within-class and between-class covariance matrices, the Fisher ratio can be written

$$F = \text{trace}\left(\mathbf{W}^{-1}\mathbf{B}\right).$$

c) In a particular speaker recognition system, a statistical model is trained for each speaker. In the model, the probability density of the N-dimensional input feature vector, $\mathbf{x}$, is modelled by the multivariate Gaussian distribution given by

$$p(\mathbf{x}) = (2\pi)^{-N/2}|\mathbf{C}|^{-1/2}\exp\left(-\frac{1}{2}(\mathbf{x}-\mathbf{m})^T\mathbf{C}^{-1}(\mathbf{x}-\mathbf{m})\right).$$

The mean $\mathbf{m}$ is defined as $\mathbf{m} = E\{\mathbf{x}\}$ and the covariance matrix $\mathbf{C}$ is given by

$$\mathbf{C} = E\{\mathbf{x}\mathbf{x}^T\} - \mathbf{m}\mathbf{m}^T$$

where $E\{\}$ denotes the expectation operator.

i) Consider such a speaker recognition system using $N = 2$ with models for 2 speakers, speaker A and speaker B. The mean feature vectors for speakers A and B are

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

respectively, and the covariance matrices for both speaker models are identity matrices, $\mathbf{I}$.

Show that, if $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, then the probability that x_1 is less that a threshold T for speaker A is given by

$$pr(x_1 < T) = \Phi(T-2)$$

$$\text{where} \quad \Phi(y) = (2\pi)^{-1/2} \int_{t=-\infty}^{y} \exp\left(-\frac{1}{2}t^2\right) dt.$$

[2]

Solution: If the covariance matrix is the identity then the two components and are independent unit variance gaussians. Therefore

$$\begin{aligned} pr(x_1 < T) &= (2\pi)^{-1/2} \int_{x=-\infty}^{T} \exp\left(-\frac{1}{2}(x-m_1)^2\right) dx \\ &= (2\pi)^{-1/2} \int_{x=-\infty}^{T-m_1} \exp\left(-\frac{1}{2}t^2\right) dt \\ &= \Phi(T-m_1) \end{aligned}$$

and $m_1 = 2$.

ii) The speaker identification system as specified in c) i) determines a test speaker to be speaker A if $x_1 < T$ and to be speaker B if $x_1 \geq T$. Find the probability of correct identification when $T = 2.5$, $T = 3$ and $T = 3.5$. [4]

It can be assumed that the prior probabilities of both speakers is 0.5. Values of the function $\Phi(T)$ can be read from Table 1.

T	0	0.5	1.0	1.5	2.0	2.5
$\Phi(T)$	0.5	0.6915	0.8413	0.9332	0.9772	0.9938

Table 1

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Solution: If the test speaker is speaker A then the probability of recognizing correctly is $\Phi(T-m_1)$. If the test speaker is speaker B then the probability of recognizing correctly is $1-\Phi(T-m_2)=\Phi(m_2-T)$.
Next, using the equal priors as given, the overall probability of recognizing correctly is $\frac{1}{2}(\Phi(T-m_1)+\Phi(m_2-T))$ giving:

$$T=2.5: \quad \frac{1}{2}(\Phi(0.5)+\Phi(1.5))=0.8124$$
$$T=3: \quad \frac{1}{2}(\Phi(1)+\Phi(1))=0.8413$$
$$T=3.5: \quad \frac{1}{2}(\Phi(1.5)+\Phi(0.5))=0.8124$$

d) A linear transformation, $\mathbf{F}$, is applied to the input feature vectors in the system of part c) to give corresponding transformed feature vectors

$$\mathbf{y}=\mathbf{F}\mathbf{x}.$$

i) Explain briefly why it can be advantageous in a speech or speaker recognition system to apply such a transformation, $\mathbf{F}$, to the input feature vectors before they are used as the input of a pattern matching process. [2]

Solution: A linear transformation can be advantageously used

- to make the within class covariance matrices approximately diagonal thus saving computation.
- to compress the useful discriminative information into a smaller number of parameters. This saves computation and can also often improves performance.

ii) Derive expressions for the mean vector $\mathbf{m}_y$ and the covariance matrix $\mathbf{C}_y$ of $\mathbf{y}$, in terms of $\mathbf{m}$ and $\mathbf{C}$. [2]

Solution: Expressions for the mean vector $\mathbf{m}_y$ and the covariance matrix $\mathbf{C}_y$ are derived as

$$\begin{aligned}\mathbf{m}_y &= E\{\mathbf{F}\mathbf{x}\}=\mathbf{F}\mathbf{m},\\ \mathbf{C}_y &= E\{\mathbf{y}\mathbf{y}^T\}-\mathbf{m}_y\mathbf{m}_y^T\\ &= E\{\mathbf{F}\mathbf{x}\mathbf{x}^T\mathbf{F}^T\}-\mathbf{m}_y\mathbf{m}_y^T\\ &= \mathbf{F}\mathbf{C}\mathbf{F}^T+\mathbf{F}\mathbf{m}\mathbf{m}^T\mathbf{F}^T-\mathbf{m}_y\mathbf{m}_y^T=\mathbf{F}\mathbf{C}\mathbf{F}^T.\end{aligned}$$

3. a) Let $V(z)$ be the system function representing a vocal tract in the z domain.

i) Write down an expression for the corresponding complex cepstrum coefficients, $c(n)$, in terms of V. [2]

ii) Write down an expression for the z transform of the complex cepstrum coefficients, $C(z)$ in terms of $c(n)$. [1]

iii) Consider the discrete-time sequence

$$x(n) = \delta(n) + \beta\delta(n-P)$$

for P being an integer constant and $|\beta| < 1$. Find the complex cepstrum of $x(n)$. [4]

You may use the power series expansion

$$\log(1-q) = -\sum_{n=1}^{\infty} \frac{q^n}{n} \quad \text{for } |q| < 1.$$

Solution:

$$c(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \log\left(V(e^{j\omega})\right) e^{j\omega n} d\omega$$

$$C(z) = \sum_{-\infty}^{\infty} c(n) z^{-n}$$

Taking z-transform of $x(n)$ gives

$$X(z) = 1 + \beta z^{-P}.$$

The z-transform of c(n) is

$$\begin{aligned} C(z) &= \log\left(X(z)\right) \\ &= \log\left(1+\beta z^{-P}\right) \\ &= -\sum_{n=1}^{\infty} \frac{(-1)^n \beta^n}{n} z^{-nP}. \end{aligned}$$

The complex cepstral coefficients having this z-transform are (by inspection)

$$c(n) = -\sum_{k=1}^{\infty} \frac{(-1)^k \beta^k}{k} \delta(n-kP).$$

b) Consider the case of the vocal tract filter comprising an all-pole filter with order p

$$V(z) = \frac{1}{A(z)}.$$

Write down general expressions for $A(z)$ in terms of the following parameters. State any assumptions made.

i) Reflection coefficients, r_k. [2]

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ii) Log area ratios, g_k. [2]

Solution:

$$A(z) = \begin{pmatrix} 1 & -1 \end{pmatrix} \prod_{k=1}^{p} \begin{pmatrix} 1 & -r_k \\ -r_k z^{-1} & z^{-1} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$g_k = \log \begin{pmatrix} 1+r_k \\ 1-r_k \end{pmatrix}$$

Assumption of reflection coefficient at the glottis $= 1$.

c) i) Explain the differences between covariance LPC and autocorrelation LPC and state their relative advantages. [3]

Solution:

The difference between covariance and autocorrelation LPC is in their choice of speech frame, F. For autocorrelation LPC analysis chose F to be infinite in extent but bounded by a windowing function, for covariance LPC analysis chose F to be a finite segment of speech: $F = s(n)$ for $0 \leq n \leq (N-1)$. Note that in a direct implementation of the covariance method, samples $s(n)$ for $-p+1 \leq n \leq 0$ are required.

<u>Autocorrelation</u>

- Requires a windowed signal giving rise to an unavoidable tradeoff between spectral resolution and time resolution.
- Requires >20 ms of data.
- Has a fast algorithm because uses inversion of toeplitz matrix.
- Guarantees a stable vocal tract filter.

<u>Covariance</u>

- No windowing required.
- Gives infinite spectral resolution.
- Requires >2 ms of data.
- Slower algorithm because F is not Toeplitz.
- Sometimes gives an unstable filter vocal tract filter.

ii) A particular signal is made from the sum three tones at 800 Hz, 1 kHz and 1.2 kHz, with relative amplitudes 1, 1 and 0.2. White noise with low relative level is added to this signal. This signal is analysed using 10th order autocorrelation and covariance LPC. In no particular order, Figure 3.1 shows the resulting filter spectra, as functions of normalised frequency.

Determine the relative amplitude in dB of the tone at 1.2 kHz. [1]

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Explain what characteristics of these spectra can be used to determine which corresponds to autocorrelation LPC and which corresponds to covariance LPC. [2]

Give an explanation for the high frequency peak observed in the Fig. 3.1(a). [1]

How would you modify the analysis in order to obtain a more accurate spectrum. [2]

Solution:
The amplitude of the 1.2 kHz tone should be

$$20\log_{10}(0.2) = -14 \text{ dB}.$$

Since covariance LPC does not have the frequency vs. time resolution trade-off associated with the analysis window needed for autocorrelation LPC, it can be deduced that the plot showing the 3 distinct peaks is obtained from covariance LPC. Note that the peak amplitudes are not correct. The high

frequency peak occurs spuriously due to a complex conjugate pole pair erroneously very close to the unit circle. This occurs in the covariance LPC because of sensitivity to the white noise. An improvement would be expected by using an increased

duration of analysis window in the case of the covariance LPC. This will reduce the estimation variance in the spectral estimate and can be tolerated because the signal is stationary.

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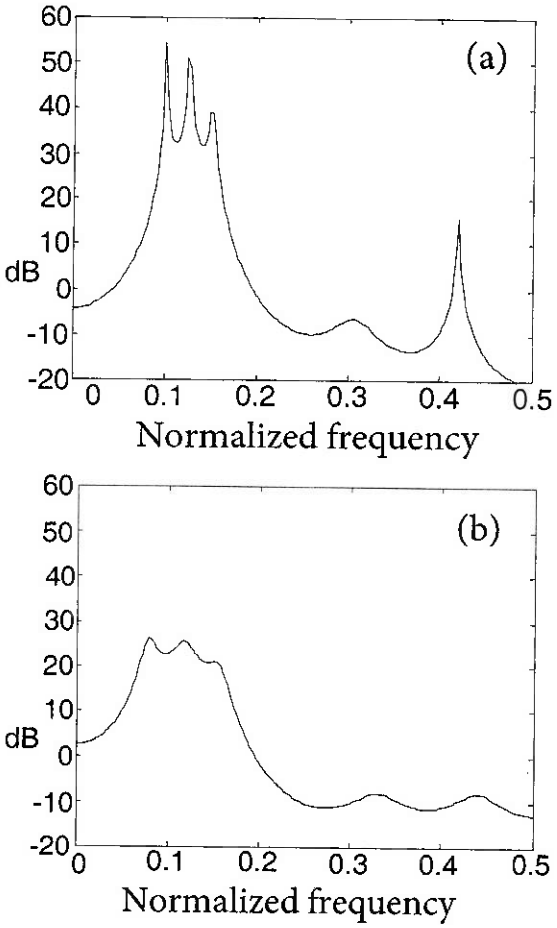


Figure 3.1: Filter spectra

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4. a) Define and explain 3 measures for the objective assessment of the quality of speech in a speech coding application. [6]

Solution:
The solution should give mathematical definitions as well as explanations of 3 such measures. Examples are given below.

- Segmental SNR: computed as the average of $10\log(E_{\text{speech}}/E_{\text{error}})$ where E_{error} is the coding error. It should be mentioned that this measure in not well correlated with subjective opinion and is not good for coding schemes that introduce delays as even a one-sample time shift causes a large change in SNR.

- Itakura Distance: computed as

$$\log\left(\left\langle\frac{P(\omega)}{Q(\omega)}\right\rangle\right)-\left\langle\log\left(\frac{P(\omega)}{Q(\omega)}\right)\right\rangle$$

where <> indicates the average over frequency.

- Itakura-Saito Distance: computed as

$$\left\langle\frac{P(\omega)}{Q(\omega)}-\log\left(\frac{P(\omega)}{Q(\omega)}\right)-1\right\rangle$$

Both the Itakura and Itakura-Saito Distances are evaluated over 20 ms frames and averaged. Both may also be calculated directly from the LPC coefficients of original and coded speech. Neither is a true distance metric since they are asymmetrical.

b) Consider a speech signal $s(n)$ and the block diagram of the main elements of a speech coder shown in Fig. 4.1.

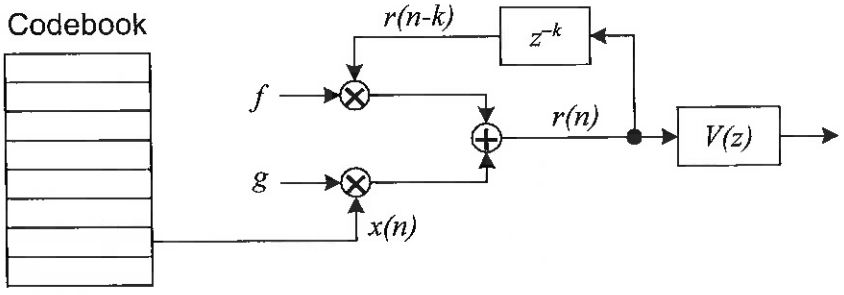


Figure 4.1: Components of a speech coder

i) Identify and explain the operation of the parts of this coder that are responsible for modelling the periodic components and the non-periodic components of the excitation signal $r(n)$. [2]

Solution:
The periodic component is modelled by the upper feedback loop that involves a delay of $z^{-}k$. The aperiodic component is modelled by selecting an entry from a fixed codebook, the selection being optimized to give the lowest re-synthesis error.

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ii) In this coder, it is decided to determine the values of k and f so as to minimize the energy function

$$E(k,f) = \sum_{n=0}^{N-1} (r(n) - r(n-k)f)^2.$$

Assume that the value of k is given and $\hat{}$ indicates the optimal value. Show that the optimum values $\hat{f}$ and the corresponding energy function $\hat{E}$ are given by

$$\hat{f}(k) = \frac{\sum_{n=0}^{N-1} r(n)r(n-k)}{\sum_{n=0}^{N-1} r^2(n-k)}$$

and

$$\hat{E}(k) = \sum_{n=0}^{N-1} r^2(n) - \frac{\left(\sum_{n=0}^{N-1} r(n)r(n-k)\right)^2}{\sum_{n=0}^{N-1} r^2(n-k)}.$$

[6]

Solution:
The optimum is found by setting the derivative of E to zero.

$$\frac{\partial E}{\partial f} = -2\sum (r(n) - fr(n-k))\, r(n-k)$$

so that

$$\hat{f} = \frac{\sum r(n)r(n-k)}{\sum r^2(n-k)}.$$

By substituting $\hat{f}$ into E we obtain

$$\begin{aligned}
\hat{E} &= \sum_{n=0}^{N-1} (r(n) - r(n-k)f)^2 \\
&= \sum r^2(n) - 2\hat{f}\sum r(n)r(n-k) + \hat{f}^2 \sum r^2(n-k) \\
&= \sum r^2(n) - 2\frac{(\sum r(n)r(n-k))^2}{\sum r^2(n-k)} + \frac{(\sum r(n)r(n-k))^2}{\sum r^2(n-k)} \\
&= \sum r^2(n) - \frac{(\sum r(n)r(n-k))^2}{\sum r^2(n-k)}.
\end{aligned}$$

iii) In a particular operation of this speech coder, the signal $r(n)$ has the following values and is zero for all other n.

$$\begin{aligned}
r(-100) &= 9 \\
r(-95) &= 4 \\
r(-50) &= 10 \\
r(-45) &= 7 \\
r(0) &= 5 \\
r(5) &= 8
\end{aligned}$$

For $N = 40$, determine $\hat{f}(k)$ and the corresponding values of $\hat{E}(k)$ for $k = 45$ and $k = 50$. [6]

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Solution:
In this particular case, there are only 2 terms that contribute to the summations in the numerator and denominator giving:

$$k = 45: \quad \hat{f}(k) = 0.714, \hat{E}(k) = 64$$
$$k = 50: \quad \hat{f}(k) = 0.711, \hat{E}(k) = 13.591.$$

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